

Final project grading consists of 10% proposal, 30% final presentation, and 60% final report.

Project Presentation Guideline. Your project presentation should provide a concise overview of your problem statement, the methods you implemented, and the evaluation of these methods. The grading rubric will be as following:

1. Problem statement (20 points):

- The problem is clearly presented and defined.
- The significance and relevance of the problem is justified.

2. Methodologies and data processing (20 points):

- The chosen machine learning algorithms are clearly explained.
- The applied data preprocessing and feature selection are discussed.

3. Implementation and evaluation (40 points):

- Results are presented and interpreted effectively.
- The evaluation metrics and their appropriateness are discussed.
- Challenges faced during implementation are discussed.
- Limitations and potential areas for improvement are mentioned.

4. Organization of presentation and slides (20 points):

- Effective use of the 10-minute time frame to include the above items.
- Clear and error-free slides including relevant visuals (e.g., charts, graphs, plots) that are useful for clear explanation.

Total Points: 100

Project Report Guideline.

The project report must be in a PDF file, preferably generated by LaTeX, and it should be at most 8 pages, including figures/tables. The font size must be 11 pt and unlimited appendices are allowed. Your report should include the following sections (the order can be changed):

1. **Title:** Include a descriptive title and also the size of your group.
2. **Abstract:** A brief overview of the report.
3. **Motivation or Problem statement:** What is the topic or goal of your project? Why do you think it is interesting or worth pursuing? Why do you think it is not trivial?
4. **Related work:** Review the relevant literature needed to properly motivate and explain the significance of the problem your work addresses.
5. **Method:** Explain the method or approach you are developing and background as much as needed (assuming the reader would have understood the topics covered in this class). A blind copy-and-paste is not allowed. Discuss the attributes of the method that warrant its use. Define all terms and variables. You should try to teach the reader as if you want them to be able to understand and implement the method for themselves. Mention any necessary data cleaning or pre/post-processing.
6. **Evaluation:** Discuss performance measures used, results, parameter tuning, design choices, etc. We hope to see quantitative, objective evaluations. Comparisons with other works is a plus. Ideally, someone should be able to reproduce your work based on what you describe. It is not necessary to show that your chosen method is superior to others. Your conclusion may indeed be that a method has some weaknesses that were not highlighted in the original publication. We encourage the use of figures to make concepts visual and concrete. Figure captions should state what the reader should conclude from the figure and they should be pretty much self contained.
7. **Conclusions:** What did you learn? What were the methods' strengths and weaknesses? What were the project's success and failures? You need not report only positive findings, you may conclude that a method was not as successful as advertised or that your data application results were not as good as anticipated due to missing data or other complications.
8. **Contribution by Group Members:** Use an additional paragraph right above references to describe the contribution from each group member, including prep work before the proposal. This paragraph together with the references won't count toward the page limit. Please make the description as specific as possible, e.g. the writing of specific sections, specific coding, parameter tuning, debugging and experiments, etc. The final writing of the statement needs agreements from all group members.
9. **Bibliography:** A list of references cited in the report.

Project Report Grading Guideline.

1. Writing Quality (15 points):

- The document is well organized and exhibits clarity in writing.
- It is easy to follow, with accurate and technically sound math notations and phrasing.
- No obvious errors or typos are present.

2. Problem Statement and Motivation (10 points):

- A brief abstract is included.
- The problem under consideration is clearly stated, motivated, and its significance is discussed.

3. Literature Review (10 points):

- Comprehensive review of existing methods is included, discussing how they relate to the current work.
- The related work is properly cited.

4. Methodologies and Data Analysis (15 points):

- The methodologies employed are clearly explained.
- The utilized dataset and its appropriateness is discussed.
- A comprehensive explanation of the used data analysis techniques is provided.

5. Evaluation (20 points):

- The experimental setup is described, including the details of appropriate machine learning libraries and tools.
- The utilized performance measures are described.

6. Conclusion (10 points):

- The key findings and lessons learned are described.
- The pros and cons of the studied method(s) are described.

7. Project Workload (20 points):

- The amount of work undertaken is deemed appropriate for the given group size.

Total Points: 100